Fire Science and Technolgy Vol. 5 No. 1, 1985.
(79～90)

IRRITATING EFFECTS OF FIRE SMOKE ON VISIBILITY

Tadahisa JIN, and Tokiyoshi YAMADA

Fire Research Institute, Fire Defence Agency,
Ministry of Home Affairs
14-1, Nakahara 3 chome, Mitaka, Tokyo 181, Japan

(Received, July 8, 1985)

ABSTRACT

The product of visibility and the smoke density as expressed by the extinction coefficient is almost constant. This product depends mainly on the brightness of the object, the illuminating light in the room, and the smoke properties. However, with irritant smoke this simple relation can apply only to low smoke densities.

Experiments were made to elucidate the effects of irritant smoke on visibility. Twelve subjects were evaluated on their visual acuity with Landolt's ring test chart in a highly irritant smoke filling room and eye blink rate was measured, as an objective indication of eye irritation.

It is concluded that the lower visibility through fire smoke is attributed to the running tears due to irritants, as well as to the extinction coefficient. Based on this new understanding, the degree of physiological and physical effect is discussed.

KEY WORDS; VISIBILITY, FIRE SMOKE, IRRITATIONS, EYE BLINK RATE, VISUAL ACUITY

1. Introduction

High-rise buildings fires have claimed steadily increasing number of victims in recent years and most of them have been claimed by smoke. In view of this fact, studies on visibility through fire smoke are of importance for safe escape. There is an increasing tendency for the results of visibility studies to be reflected in the establishment of internal building materials restrictions and smoke control regulations, especially for high-rise buildings.

Although a number of workers have been reported on visibility through smoke [1, 2, 3, 4], little work still has been done to estimate the factors affecting visibility which may arise in an actual building fire; for example, the effects of smoke properties, the brightness of objects, and biological effect of smoke on human eyes.

The present paper describes experimental results on irritant effects of smoke on visibility.

2. Smoke Density and Visibility

Visibility of signs through fire smoke can be expressed approximately by the following equation [5] (see Appendix)

$$V \fallingdotseq \frac{1}{C_s} \ln\frac{B_{EO}}{\delta_c KL} \qquad (1)$$

where V = visibility of signs at the obscuration

threshold (m)

C_S = smoke density expressed by the extinction coefficient (1/m)
* (hereafter, smoke density will be expressed by the extinction coefficient in 1/m)

B_{EO} = brightness of signs (cd/m^2)

δ_c = contrast threshold of signs in smoke at the obscuration threshold. (0.01 – 0.05)

K = σ_s / C_s. (0.4 – 1.0)

and C_S = $\sigma_s + \sigma_{ab}$. (σ_s = scattering coefficient; σ_{ab} = absorption coefficient)

L = $1/\pi$ of mean illuminance of illuminating light from all directions in smoke ($1\,m/m^2$)

For placard-type (reflecting) signs Equation (1) can be modified to

$$V \fallingdotseq \frac{1}{C_s} \ln\frac{\alpha}{\delta_c K} \qquad (2)$$

where α = reflectance of sign.

The signs in a smoke-filled chamber were observed from outside through a glass window. The results are shown in Fig. 1.

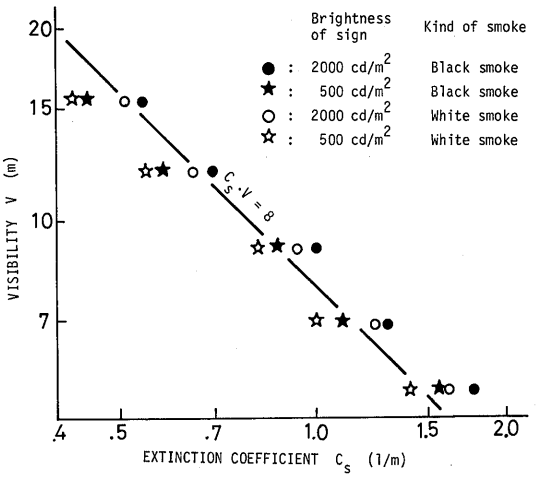


Figure 1. Relation between the visibility of light-emitting signs at the obscurity threshold and smoke density (Extinction coefficient)

Fig. 1 shows the relation between the visibility of light-emitting signs at the obscuration threshold and the density of smoldering smoke (white) or flaming smoke (black). In the range of the visibility of 5-15m, the product of the visibility (V) at the

* The extinction coefficient (Cs) can be obtained by the following equation:

$$C_s = \frac{2.3}{\ell} \log_{10} \frac{I_0}{I} \text{ or } I = I_0 e^{-C_s \ell}$$

where Io = the intensity of incident light
I = the light intensity through smoke
ℓ = light path length (m)

obscuration threshold and the smoke density (C_S) is almost constant.

The visibility in black smoke is somewhat better than in white smoke of the same density; this remarkable difference in visibility is not recognized among smokes from various materials. For reflecting signs, the product of the visibility and smoke density is almost constant, too. The product depends mainly on the reflectance of the sign and the brightness of illuminating light.

The visibility at the obscuration threshold of signs is found to be,

(5-10) $/C_s$ (m) for a light-emitting signs, and,

(2-4) $/C_s$ (m) for a reflecting sign

The visibility of other objects such as walls, floor, doors, stairway, etc. in an underground arcades or a long corridor varies depending on the surrounding relactance and contrast condition, however, the minimum value for reflecting signs may be applicable.

3. Visibility of Signs through Irritant Smoke and Walking Speed

A 20-meter long corridor was filled with smoke corresponding to an early stage of fire; a highly irritant white smoke was produced by burning wood cribs with narrow spacing between the sticks, and a less irritant black smoke was produced by burning kerosene. The subjects were instructed to walk into the corridor from one end, or to record the places at

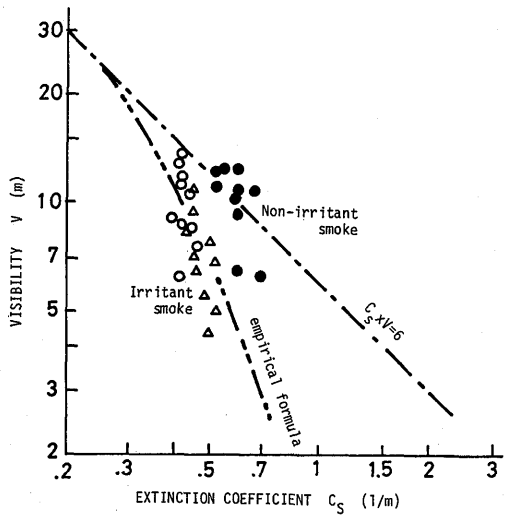


Figure 2. Visibility of the FIRE EXIT sign at the legible threshold of the words in irritant or non-irritant smokes.

where they saw a lighted FIRE EXIT sign at another end, or to read the words on the signs. (ref. 5) For the obscuration threshold of the sign, the following relation can be found $C_s \cdot V \fallingdotseq$ constant. However, for visibility at the legible threshold of words, this relation can only apply to non-irritant smoke as shown in Fig. 2.

In Fig. 2, the visibility in irritant smoke decreases sharply at the smoke density exceeding a certain level. In the thick irritant smoke, the subjects could not keep their eyes open for a long time, and tears ran so heavily that they could not see the words on signs.

Even in this case, when the exit signs are so simple or familiar with occupants to understand at a glance, this irritant effect of smoke may not cause a trouble so much to recognize

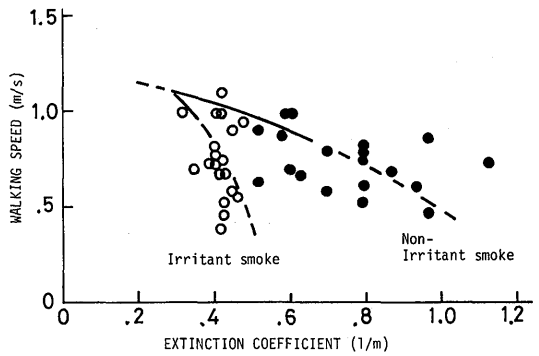


Figure 3. Walking speed in fire smoke.

the exits.

The smoke irritation reduces the visibility of evacuees moreover and subsequently there will be a possibility to give them needless unrests or panic. The smoke hazards of notice are found not only in such psychological aspects [6, 7], but also in their actions, especially walking speed.

Fig.3 shows an example data of the experiment of walking speed in the smoke conducted before. Both smoke density and irritation affect the walking speed apparently. This figure shows that the walking speed in the none irritant smoke decreases gradually as increasing the smoke density. However, in the irritant smoke the speed decreases rapidly in the region over certain smoke density level as same as the visibility dose. From the observation of this experiment, this sharp drop of walking speed is explained by the subjects' movement; they could not keep their eyes open, then they walked inevitably zigzag or step by step along the side wall.

4. Further Laboratory Study of Visibility through Irritant Smoke

The results above mentioned led us to the hypothesis that the irritation of smoke could be one of the major responsible parts of the visibility in case of fire. Therefore, the aim of the present study is to examine to what extent this irritation is responsible for the visibility.

In a preceding experiment, a japanese fire exit sign written in Chinese character "非常口" was used for a test chart of visibility, however it was not necessarily enough to generalize the effect of irritant smoke on visibility.

In this time, further laboratory studies were conducted with universal Landolt's ring test chart in the high irritant smoke-filling room. The relation between the visual acuity as to the visibility and the eye blink rate were also examined in this experiment.

The eye blink rate was adopted as an index of eye annoyance and of tear flow rate. [8]

4.1 Test Method

The experiments were made in a room of 18 m² as shown in Fig.4. This room was filled with the high irritant white smoke produced by smoldering wood chips in an electric furnace. The smoke density increased at the rate of about 0.05 (1/m • sec).

Twelve subjects including one female were 20 to 30 years old and had normal visual acuity over 1.0. They were exposed to the irritant smoke

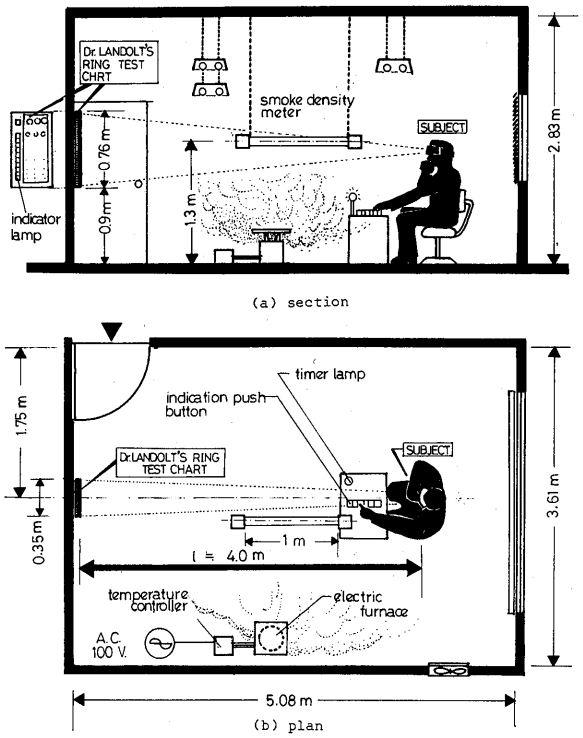


Figure 4. Layout of the smoke study laboratory: (a) section, (b) plan

under two conditions mutually. Under one condition, subjects wore the well-sealed goggle for protecting their eyes from the irritant smoke, and under another wore the none-sealed goggle. Both goggles were made for skiing use and almost the same products except that one of them was not sealed, moreover numbers of hole were made on the sides of that frame to let the irritant smoke flow in (hereafter this case will be expressed "no goggle" .or "without goggle".) Under both conditions, the subjects covered their noses with a towel to prevent the irritant smoke from effecting on nose. About 75 % of the smoke density in breathing is removed by using this towel. [9]

The front part of this goggle was colored in light yellow and the transmittance was measured as 65% with a luminance meter for spectral luminous efficiency. This transmittance will affect the visibility physically, however, there will be little problems to assess the irritant effect of smoke on visibility because this effect will be canceled each other.

The visibility was presented by the visual acuity. Each subject entered in the room every one or two minutes with goggle and sat on a chair located 4m in just front of Landolt's ring test chart. The indication push button was set on the desk. As a subject pushes the button corresponding to a visible point, the lamp of that point on the test chart is lit and the indicated value is recorded out of the room. Thus after 30 sec. waiting, the subject pushed the button corresponding to the legible threshold, then the visual acuity value was recorded in another room. As pushed the button, subject went out of the room, wore another goggle, then reentered in the room and the same routines were done repeatedly for about 15 min. on each subject. The extinction coefficient of smoke measured continuously at 1.3 m hight between visual test chart and subject. For this 15 minutes duration, this value was increased slowly.

The eye blink rate was measured during first 30 sec. in each routine with small strain gage (3 mm x 5 mm)

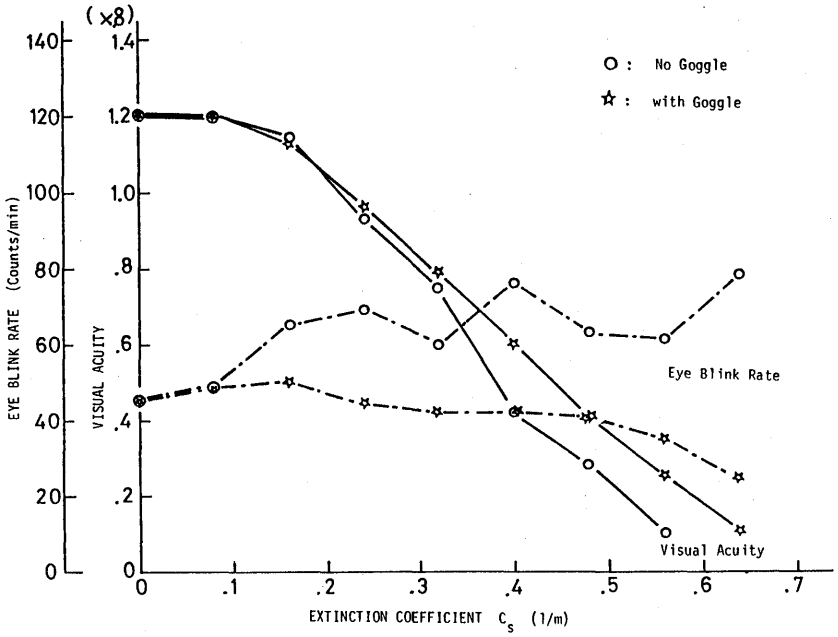


Figure 5. Relation between the visual acuity, the eye blink rate and the smoke density (extinction coefficient)

developed for a medical inspection. This gage was fixed on the skin closed to a tail of the eye. The outputs of the eye blink were telemetered and could be recognized as remarkable sharp waves.

An inside of the laboratory was illuminated under the same condition as usual. The normal illuminance on the surface of Landolt's ring test chart was about 640 lx., and the luminance of that chart was about 150 cd $/ m^2$ of spectral luminous efficiency measured at the subject's point with luminance meter. These values were not so varied within about 5 % at the most under the smoke filled condition.

This result is explained in terms of the more scattering and less absorbing effects through the white smoke. [10]

4.2 Result

As the smoke density is increased, the average visual acuity of twelve subjects decreases shown in Fig. 5. The relationship between the visual acuity and the extinction coefficient is almost linear in the region of over 0.2 (1/m) of extinction coefficient. This decrease of visual acuity is caused by the physical characters of smoke absorption and decrease of apparent area due to smoke particles for example, and by the chemical properties. The irritative composite of smoke induces the obscuration with the heavily running tears as above mentioned.

The decrease rate of visual acuity to the smoke density without goggle is about 15% higher than that with gog-

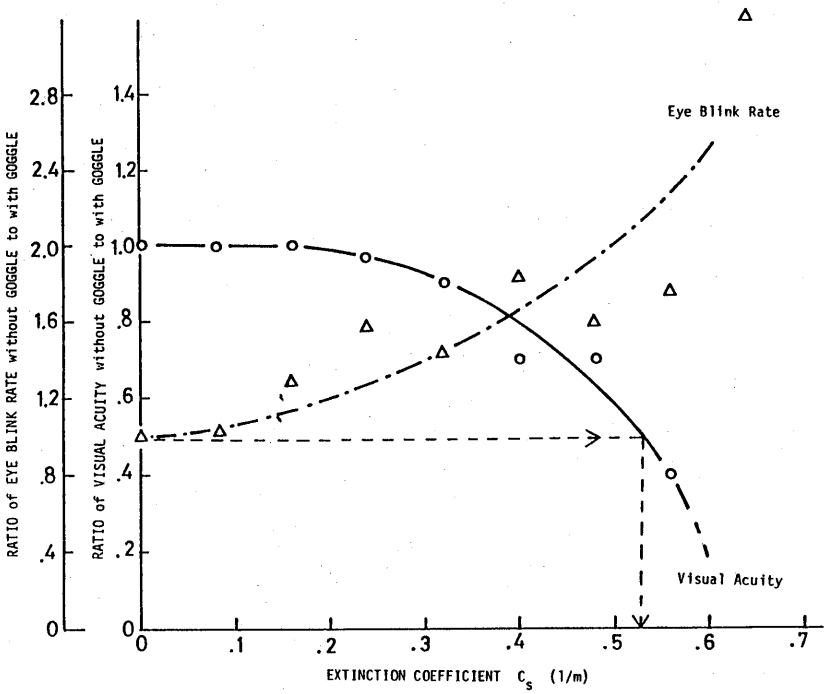


Figure 6. Effect of with goggle on visual acuity and eye blink rate due to irritating smoke; the effect is expressed with the ratio without goggle to with goggle.

gle in the range of over 0.2 (1/m). These results show that the effects of irritant smoke are not constant but increase as the smoke density increases.

Another result of the relationship between the eye blink rate and the smoke density is shown in Fig. 5 by the dash dotted line. The eye blink rate increases gradually with increasing the smoke density in case of no goggle. In another case, the rate is almost constant.

The experiments under the higher density smoke were conducted after the several experiments under the lower density smoke continuously on each subject. Presumably continuous exposures of smoke fitted the subjects to the irritant condition physiologically, however this effect could not be eliminated in this study.

4.3 Discussion

Fig. 6 indicates the ratios of the visual acuity and the eye blink rate without goggle to those with goggle. The visual acuity ratios are almost constant in the region of under 0.25 (1/m) smoke density and drop rapidly over 0.25 (1/m).

On the other hand, the eye blink rate ratios are increasing gradually in the region of lower smoke density and rise higher by the smoke density increases. These results suggest that the use of goggle in case of evacuation through thick fire smoke is one of the useful methods to pre-

vent the visibility decrease caused by the irritant smoke.

The relationship between the visual acuity and the eye blink rate is shown in Fig. 7. Under condition as the eye blink rate ratio ranging from about 1.0 to 1.5, almost all subject felt a little pain in their eyes, though the visual acuity decrease was not observed in this region. With the region ranging 1.5 to 2.0, the visual acuity ratio decreases rapidly and over 2.0, presumably nothing can be seen beyond 4 m depth due to the heavily running tears.

Weber-Fechner's Law gives the relationship between an impact on a human sensitivity and its response; the response of human sensitivity is logarithmically proportional to the impact intensity.

This well known theory is applied to the series of experiments, that is,

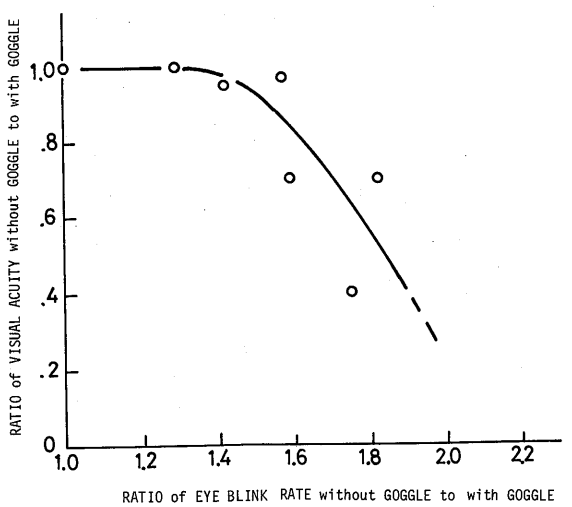


Figure 7. Relation between the eye blink rate and the visual acuity. Each value is expressed as the same ratio presented in Fig. 6.

$S = A - B \log C_s$, where S is the relative visual acuity as the response, Cs the extinction coefficient as the impact and A, B is an experimental constant. The relative visual acuity S and Cs data are plotted on a semi-log chart as shown in Fig. 8. in the region of Cs over 0.25 (1/m). The plotted data in this figure indicate approximately linear relationship, then the decrease of visual acuity due to the smoke irritant effect is expressed,

$$S = 0.133 - 1.47 \log Cs \qquad (3)$$

The drop of visual acuity through smoke seems to be caused by two factors mainly. One is the apparent decrease of the visual acuity due to the physical effect of smoke particles intercepting the objects. Another is the physiological irritant effect of smoke. Thus the visibility through smoke is expressed by the next approximations.

(in none irritant region)

$$V_1 = C/Cs \quad (0.1 \leqq Cs < 0.25) \qquad (4)$$

(in irritant region)

$$V_2 = C/Cs. \quad (0.133 - 1.47 \log Cs) \quad (Cs \geqq 0.25) \qquad (5)$$

Above two equations are not applicable in the region where the smoke density is extremely thin ($Cs < 0.1$) and high ($V_2 < 0$).

Experimental constant of C in the equations depends on the several experimental conditions; the brightness of objects and its surroundings, the

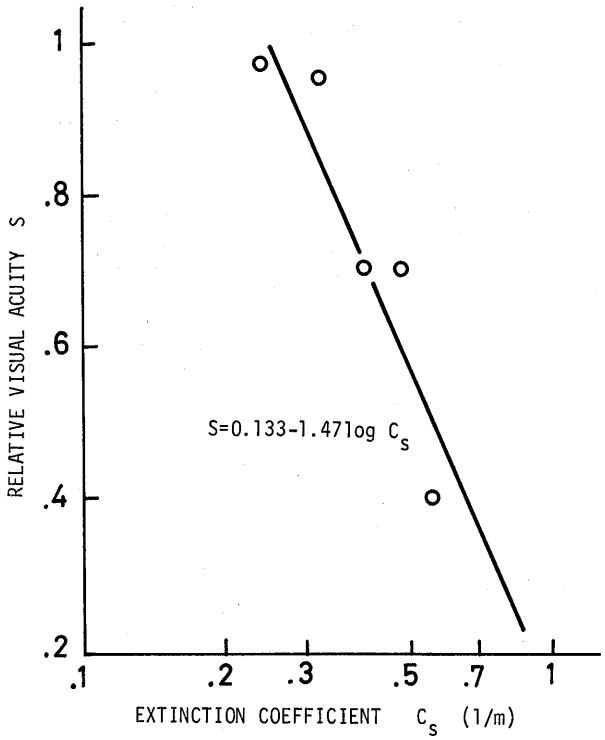


Figure 8. Decrease of relative visual acuity due to smoke irritant effect.

contrast between them and the kinds of contents on the objects.

The obtained relationships of Eq. (4) and (5) are applied to the series of experiments on the visibility through the smoke as mentioned in section 2 & 3, and then the results are presented in Fig. 2 with dash two dotted line. The theoretical curves approximately agree with the experimental data when the constant value is set to be 6.0. The visibility through other kinds of smoke besides the irritant white smoke from the smoldering wood may be considered to vary in the intermediate region between the equation (4) and (5).

5. Conclusion and Recommendation

Through many experimental studies, the following results were obtained.

(1) The visibility through thin smoke depends on the physical obscuration effect mainly, however, when the smoke is relatively thick, the physiological irritant effect become the dominant factor of visibility in place of the physical effect.

(2) In accordance with Weber－Fechner' s Law, the visibility through smoke decreases as increasing the smoke irritant in the region where the eye blink rate becomes over 1.5 times larger than that of ordinary state.

For safety evacuation through fire smoke, the use of goggle is available. This time, the irritant compound was not a major subject study, however, the relation between chemical compounds and the irritant effect will be revealed through other experiments in future.

References

[1] F. E. T. Kingman: J. Appl. Chem., 3, 463, (1953).
[2] D. J. Rasbash: Fire, 59, 735, 175, (1966).
[3] H. L. Malhotra: Fire Research Note, 651, (1967).
[4] J. A. Bono: UL Bull. of Research, 56, 146, (1965).
[5] T. Jin: J. of Fire & Flammability, 135, Vol. 9, (1978).
[6] T. Jin: J. of Fire & Flammability, 130, Vol. 12, (1981).
[7] T. Jin: Bull. of Japanese Assoc.

of Fire Sci. & Eng., 1, Vol. 32 No. 2 (1982).
[8] A. Weber, T. Fischer, and E. Grandjean: Int. Archives of Occupational and Environmental Health 43, 183–193 (1979).
[9] T. Jin: Kasai (J. of Japanese Assoc. of Fire Sci & Eng.), 26, No. 134 (1981).
[10] T. Jin: Report of Fire Research Institute of Japan, 40, 6, (1975).

[APPENDIX]

The brightness of the sign and the background are, respectively, B_{EO} (cd/m^2) and B_{BO} (cd/m^2) as shown in Fig. A−1. The sign in smoke can not be distinguished from the background, when it is V (m) away from the subject's eye. The intensities of light at the subject's eye from the sign and the background are, respectively,

$$B_{E'} = \frac{B_{EO} S e^{-C_s V}}{V^2} \qquad B_{B'} = \frac{B_{BO} S e^{-C_s V}}{V^2}$$

where, S : the area of a sign (m^2)
C_s : smoke density represented by an extinction coefficient.

Next, we consider the intensity of light which is caused by the scattering of the illuminating light usually existing in corridor or a room by the smoke particles floating in the conical space between the sign and the subject's eye (see Fig. A−1). The smoke particles contained in a small volume $S(X/V)^2 dX$ scatter illuminating light in every direction and its intensity must be proportional to that

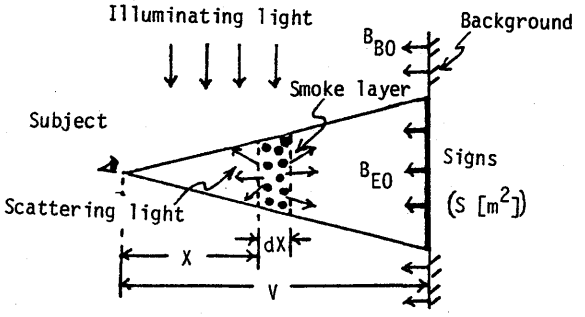


Figure A−1. Scattering of illuminating light by smoke particles.

of the incident light.

If a parallel beam of light, the intensity of luminous flux being L, illuminates this smoke layer, the scattered light in all directions is equal to $\sigma_s LS(X/V)^2 dX$.

Where, σ_s is the light scattering coefficient, that is the total luminous flux of light scattered in all directions when a parallel light beam having unit luminous flux per unit area is thrown upon unit volume of smoke. The mean intensity of the scattered light reaching the subject's eye dB′ is obtained as follows :

$$dB' = \frac{\delta_s LS(X/V)^2 dX e^{-C_s X}}{4\pi X^2}$$

$$= \left(\frac{1}{4\pi}\right)\delta_s L\left(\frac{S}{V^2}\right)e^{-C_s X}dX$$

When the lights are incident upon the smoke layer from all directions with the same intensity, the intensity of the scattered light reaching eyes dB″ is obtained by multiplying the value of the above equation by 4π i. e.,

$$dB'' = \delta_s L\left(\frac{S}{V}\right)e^{-C_s X}dX$$

As the value of L in this equation,

we can use the mean illuminance (E) multiplied by $1/\pi$ where 'mean' means the mean value of illuminance measured in all directions using an illuminometer having such angular response as obeying the cosine law of illumination.

$$L = \frac{E}{\pi}(cd/m^2)$$

L has the dimension of the brightness. As the same phenomenon occurs in all the parts between the subject's eyes and the sign, the intensity of the total scattered light reaching the eyes B″ is,

$$B'' = \int dB'' = \int \delta_s L\left(\frac{S}{V^2}\right)e^{-C_s X}dX$$

$$= \left(\frac{\delta_s}{C_s}\right)L\left(\frac{S}{V^2}\right)(1-e^{-C_s V})$$

The scattered light having the intensity B″ enters the eyes together with the lights from the sign and the background. Then, the intensity of the total light including scattered light from the sign and the background, respectively, B_E and B_B which reaches the subject's eyes is,

$$B_E = B_{EO}\left(\frac{S}{V^2}\right)e^{-C_s V}+\left(\frac{\delta_s}{C_s}\right)L\left(\frac{S}{V^2}\right)(1-e^{-C_s V})$$

$$B_B = B_{BO}\left(\frac{S}{V^2}\right)e^{-C_s V}+\left(\frac{\delta_s}{C_s}\right)L\left(\frac{S}{V^2}\right)(1-e^{-C_s V})$$

Assuming $B_{EO} > B_{BO}$, the contrast threshold δc is

$$\delta_c = \frac{B_E - B_B}{B_B}$$

$$= \frac{(B_{EO}-B_{BO})e^{-C_s V}}{B_{BO}e^{-C_s V}+kL(1-e^{-C_s V})} \qquad (A-1)$$

where $k = \sigma s / Cs$ and the following equation is generally satisfied between Cs and σ s:

$$C_s = \sigma_s + \sigma_{ab}$$

(σ_{ab} = absorption coefficient)

The equation (A-1) is rearranged based on CsV:

$$C_s V = \ln\frac{\delta_c kL-(1+\delta_c)B_{BO}+B_{EO}}{\delta_c kL} \qquad (A-2)$$

When the brightness of the sign B_{EO} is sufficiently larger than B_{BO} and δ_c kL, the equation (A-2) is,

$$C_s V \fallingdotseq \ln\frac{B_{EO}}{\delta_c kL}$$

The visibility of the sign with a certain smoke density increases almost linearly, following logarithmically proportional increase of the brightness of the sign.

The brightness of a reflecting sign B_{EO} is determined based on the brightness L of incident illuminating lights in smoke and the reflectance ρ of the sign: that is,

$$B_{EO} = \rho L \quad \text{or} \quad \rho = \frac{B_{EO}}{L} \qquad (A-3)$$

Inserting the equation (A-3) into the equation (A-2),

$$C_s V = \ln\frac{\delta_c kL+\rho L}{\delta_c kL} = \ln\frac{\delta_c k+\rho}{\delta_c k}$$

With a practical reflecting sign, $p >> \delta_c k$;

$$C_s V \fallingdotseq \ln\frac{\rho}{\delta_c k}$$